Genetics

Ingyu Bahng

June 2026

Contents

1 Meiosis and Sexual Life Cycles

1.1 Sexual Life Cycles

The transmission of heritable traits from one generation to the next constitutes **heredity**, a process through which genetic information is passed across lineages. Offspring exhibit inherited similarities yet also demonstrate **variation**, reflecting differences in the specific genetic information transmitted. The discipline of **genetics** encompasses the study of both hereditary mechanisms and variation. Genetic information is transmitted via **genes**, discrete DNA segments that encode specific functional products and serve as the fundamental units of heredity.

Core Concept 1.1: Organization of Chromosomes

Human **somatic cells** (*non-reproductive cells*) are characterized by their possession of two complete chromosome sets—one of maternal origin and one of paternal origin—yielding a total of 46 chromosomes. Such cells, containing two chromosome sets ($2n$), are designated **diploid**. *It should be noted that the diploid chromosome number exhibits considerable variation across species.* The chromosomal composition of a human somatic cell can be visualized through a **karyotype**, as illustrated to the right.

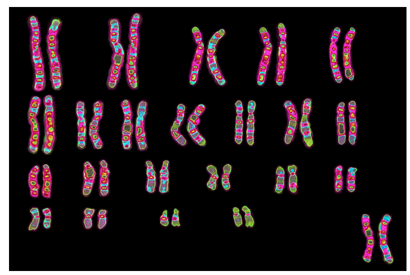


Figure 1: Human karyotype

As demonstrated Figure 1, paired chromosomes share identical length, centromere position, and gene arrangement. These matched pairs are termed **homologous chromosomes**. Although both members of a homologous pair carry genes governing the same inherited traits, the specific allelic variants may differ between these **homologs**. This pattern of homology characterizes 22 of the 23 chromosome pairs in humans. The twenty-third pair comprises the **sex chromosomes** (*Core Concept 3.2*), which typically manifest as XX in females and XY in males, while the remaining 22 pairs are collectively designated **autosomes**.

Core Concept 1.2: Sexual vs Asexual Reproduction

Sexual reproduction proceeds through the transmission of genetic material via specialized reproductive cells termed **gametes**. These gametes are **haploid** (n), containing a single chromosome set, and arise through the specialized cell division process of **meiosis** (*Core Concept 1.4*). Offspring formation occurs when male and female gametes—sperm and egg, respectively—undergo fusion through **fertilization**, thereby reconstituting the diploid ($2n$) chromosome complement.

This reproductive mode contrasts fundamentally with **asexual reproduction**, wherein offspring derive from a single parent organism through **mitosis** (**??**), yielding genetically identical **clones**.

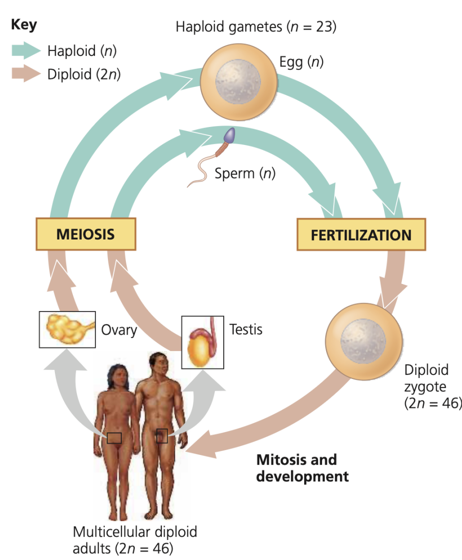


Figure 2: Human sexual life cycle

Core Concept 1.3: Sexual Life Cycles

The temporal sequence of reproductive events described in Core Concept 1.2 constitutes a **sexual life cycle**. Although the human life cycle exemplifies this pattern, considerable diversity in sexual life cycle organization exists among organisms.

Plants and certain algae exhibit **alternation of generations**, a life cycle characterized by multicellular diploid and haploid stages. The diploid generation, termed the **sporophyte**, undergoes meiosis to generate haploid **spores**. In contrast to gametes, spores do not undergo fusion but rather proliferate mitotically to establish a multicellular haploid **gametophyte**. Subsequently, the gametophyte produces gametes through mitosis. Fertilization of these gametes yields a diploid zygote that develops into the succeeding sporophyte generation. This cyclical pattern—wherein sporophytes generate gametophytes, which in turn produce sporophytes—constitutes the *alternation between generations.*

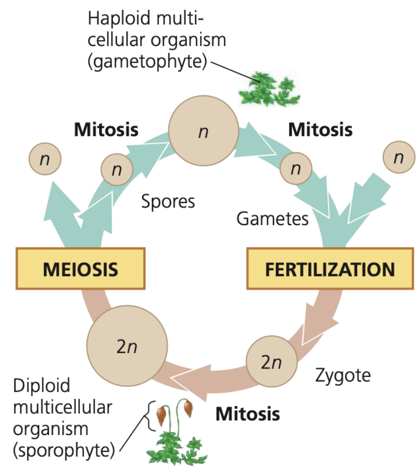


Figure 3: Alternation of generations sexual life cycle

Most fungi and certain protists, including some algal species, demonstrate a life cycle in which **the diploid phase is restricted to the unicellular zygote stage**. Following gametic fusion to form a diploid zygote, **meiosis proceeds immediately**, generating haploid cells rather than permitting development of a multicellular diploid organism. These haploid cells subsequently undergo mitotic divisions, establishing either unicellular populations or multicellular haploid organisms. Gamete production then occurs mitotically within the haploid organism, thereby completing the reproductive cycle.

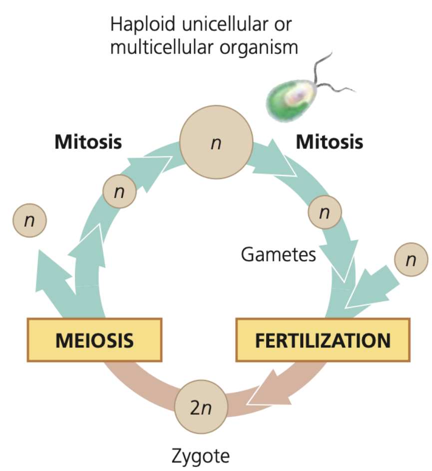


Figure 4: Life cycle with zygotic meiosis; *characteristic of most fungi and some protists*

1.2 Meiosis

Core Concept 1.4: Meiosis

Meiosis constitutes the specialized cell division process through which **germ cells** within the gonads—ovaries in females and testes in males—generate gametes. Similar to **mitosis** (**??**), meiosis is preceded by an interphase period that encompasses S phase, during which chromosomal replication occurs. Meiosis diverges from mitosis, however, in comprising two successive division events—meiosis I (*Core Concept 1.5*) and meiosis II (*Core Concept 1.6*)—that collectively yield four haploid daughter cells.

Each meiotic division encompasses the canonical phases of prophase, metaphase, anaphase, and telophase, with cytokinesis ensuing. These phases are distinguished by Roman numeral designations (*I or II*) to denote their occurrence within either meiosis I or meiosis II.

Core Concept 1.5: Meiosis I

(1) **Prophase I:** Centrosome separation initiates spindle apparatus assembly, concurrent with nuclear envelope disintegration and kinetochore formation. During this phase, **crossing over** (*Core Concept 1.7*) occurs generating cytologically visible structures termed **chiasmata**.

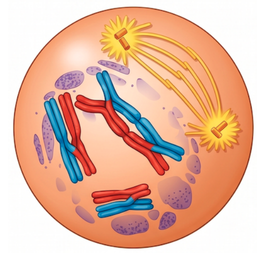


Figure 5: Prophase I

(2) **Metaphase I: Homologous chromosome** pairs align at the metaphase plate, with each pair orienting independently of all other pairs, thereby facilitating **independent assortment**.

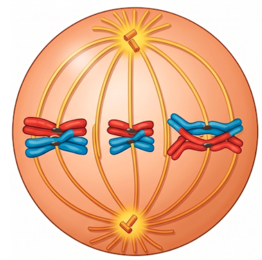


Figure 6: Metaphase I

(3) **Anaphase I:** Cleavage of cohesin proteins along chromosome arms permits homologous chromosome separation, while sister chromatids maintain attachment at their centromeric regions.

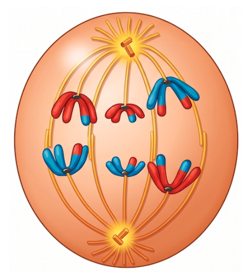


Figure 7: Anaphase I

(4) **Telophase I & Cytokinesis:** Cleavage furrow formation in animal cells partitions the cytoplasm, yielding two daughter cells, each possessing a haploid complement of duplicated chromosomes.

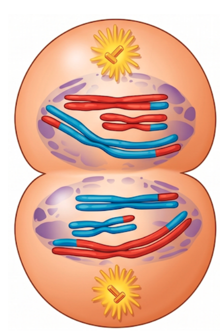


Figure 8: Telophase I

Core Concept 1.6: Meiosis II

(1) **Prophase II:** Spindle apparatus assembly proceeds, nuclear envelope disintegration occurs if reformation had ensued following meiosis I, and kinetochores establish connections with spindle microtubules.

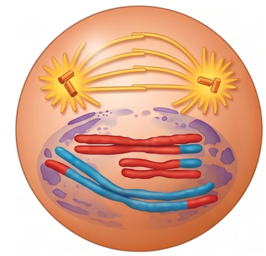


Figure 9: Prophase II

(2) **Metaphase II:** Chromosomes achieve metaphase plate alignment. Due to crossing over in meiosis I, the two sister chromatids of each chromosome are no longer genetically identical. Sister chromatid kinetochores form attachments to microtubules emanating from opposing spindle poles.

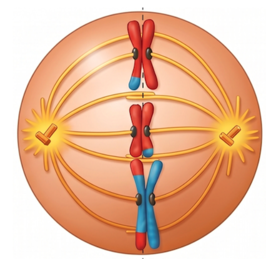


Figure 10: Metaphase II

(3) **Anaphase II:** Cleavage of centromeric cohesin proteins liberates sister chromatids, enabling their poleward migration. Each segregated chromatid now constitutes an individual chromosome.

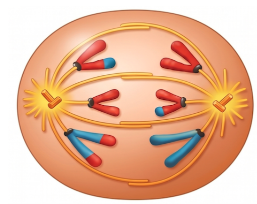


Figure 11: Anaphase II

(4) **Telophase II & Cytokinesis:** Nuclear envelope reformation encompasses each chromosome set, which initiates decondensation. Cytokinesis subsequently ensues.

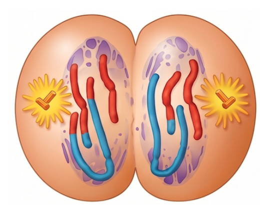


Figure 12: Telophase II

Note that the events depicted above occur in both daughter cells generated during meiosis I.

The complete meiotic process yields four daughter cells, each harboring a haploid complement of unduplicated chromosomes that are genetically distinct both from one another and from the parental cell.

Core Concept 1.7: Crossing Over

A fundamental distinction between meiosis and mitosis resides in the occurrence of **crossing over** during prophase I of meiosis. This process entails the reciprocal exchange of homologous DNA segments between nonsister chromatids of homologous chromosome pairs.

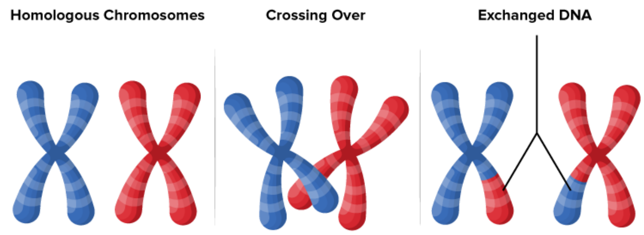


Figure 13: Crossing over

The cytologically visible sites of crossing over are designated **chiasmata**. Proper chromosomal alignment and segregation during meiosis I typically requires at least one crossover event per homologous pair, as chiasmata function to maintain homolog association until their disjunction during anaphase I.

Core Concept 1.8: Genetic Variation

Meiosis and sexual reproduction generate substantial genetic variation through three distinct mechanisms.

(a) **Independent assortment of chromosomes** describes the random orientation of **homologous chromosome** pairs at the metaphase plate during **metaphase I** of meiosis.

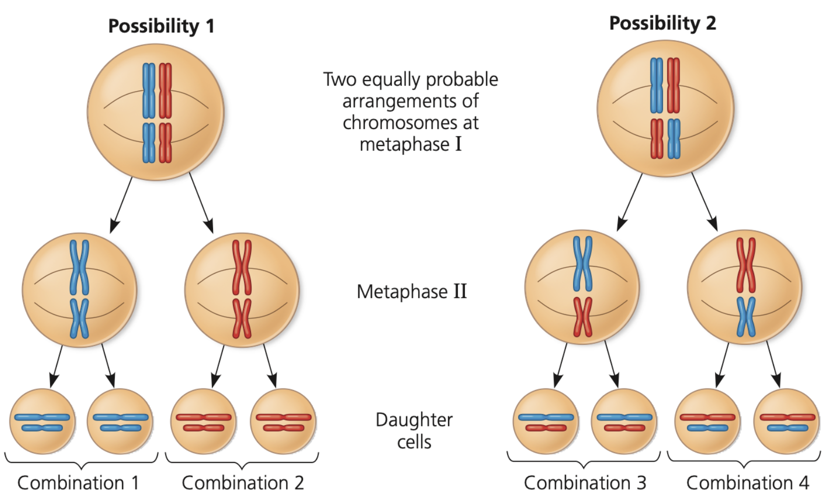


Figure 14: Independent assortment of chromosomes

Each homologous pair may orient such that either the maternal or paternal homolog faces a given spindle pole. Consequently, a 50% probability exists that any daughter cell resulting from meiosis I will receive the maternal chromosome of a given pair, with an equivalent probability for paternal chromosome inheritance.

Given that each of the 23 homologous chromosome pairs in humans assorts independently, this mechanism alone generates 2^{23} (*exceeding 8 million*) possible combinations of maternal and paternal chromosomes within a single gamete—a figure that precedes consideration of the subsequent two mechanisms.

(b) The **random nature of fertilization** amplifies genetic variation. Because any of the 2^{23} possible chromosomal combinations in a male gamete may unite with any of the 2^{23} combinations in a female gamete, the resultant zygote may exhibit 2^{46} (*exceeding 70 trillion*) possible configurations of maternal and paternal chromosomes.

(c) **Crossing over** during prophase I of meiosis I generates **recombinant chromosomes** that exhibit genetic configurations distinct from the parental maternal and paternal homologs. Given that crossing over may occur at numerous positions along each homologous pair and may transpire multiple times per pair, the spectrum of possible genetic combinations expands dramatically.

2 Mendelian Genetics

2.1 Inheritance and Mendel's Peas

Core Concept 2.1: Mendel's Experiments

Beginning around 1857, Gregor Mendel conducted systematic experiments on plant species in Brno, focusing on traits exhibiting discrete alternative forms. His experimental design employed **true-breeding** plants—lines producing identical offspring through repeated self-pollination—as the **P generation** (*parental generation*). **Hybridization** between contrasting true-breeding varieties yielded the **F1 generation**, which upon self-pollination produced the **F2 generation**. Analysis of phenotypic ratios across these generations enabled Mendel to deduce the fundamental principles of inheritance.

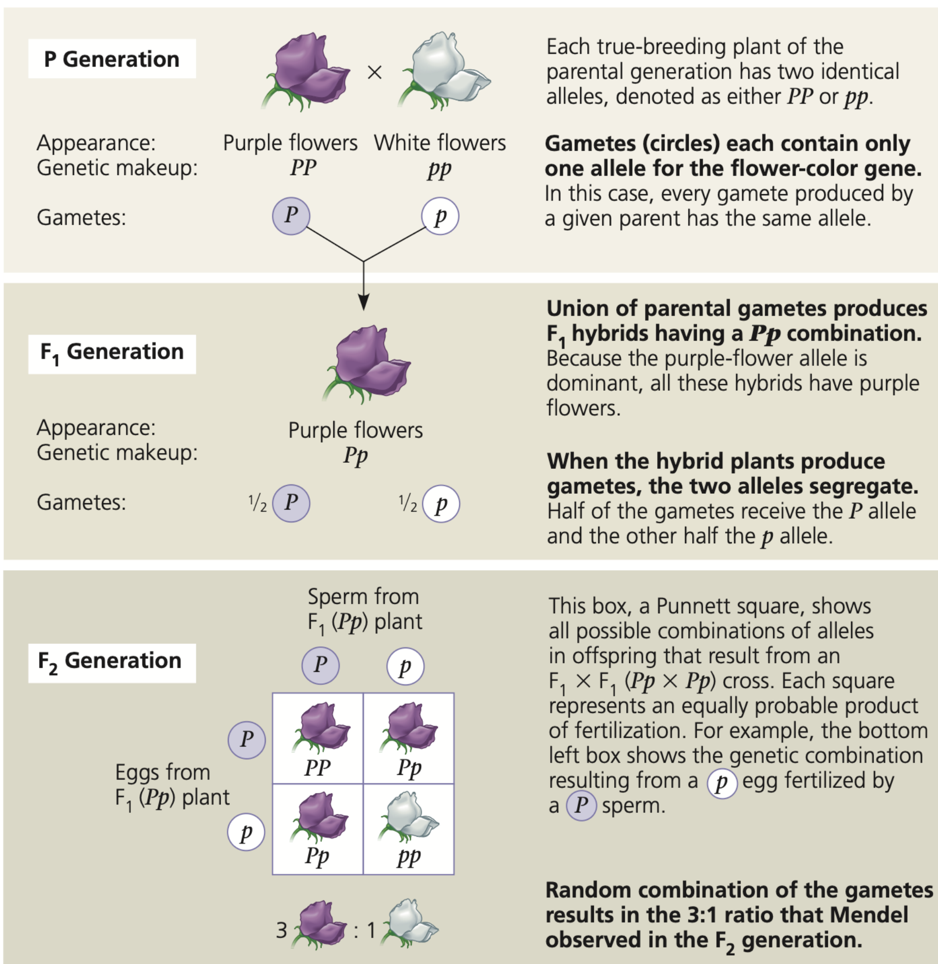


Figure 15: Mendel's experiments examining flower color

The prevailing *blending hypothesis* predicted that parental traits would blend uniformly in offspring, producing intermediate phenotypes that would persist across generations. Mendel's crosses of true-breeding purple- and white-flowered plants contradicted this model: F1 offspring displayed exclusively purple flowers rather than an intermediate color, and white flowers reappeared in the F2 generation at an approximate 3:1 ratio (*purple:white*). These results established that hereditary factors are transmitted as discrete units (*now recognized as genes*), in which **dominant** alleles mask the phenotypic expression of **recessive** alleles.

Core Concept 2.2: Mendel's Model of Inheritance

Mendel's model of inheritance rests on four foundational principles.

(a) Alternative versions of genes (*alleles*) account for variations in inherited characters.

(b) For each character, an organism inherits two **alleles** of a gene, one from each parent.

(c) When alleles at a given locus differ, the **dominant** allele (*denoted by uppercase notation, e.g., P*) determines the phenotype, whereas the **recessive** allele (*lowercase notation, e.g., p*) has no phenotypic effect in the presence of the dominant allele.

(d) The two alleles for a heritable character segregate during gamete formation, such that each gamete receives only one allele (*the law of segregation; Core Concept 2.4*).

Core Concept 2.3: Phenotypes & Genotypes

Organisms possessing two identical alleles for a gene are **homozygous** (*PP or pp*), whereas those with two different alleles are **heterozygous** (*Pp*).

Because different allelic combinations can produce identical observable traits, appearance does not necessarily reveal genetic composition. We therefore distinguish between an organism's **phenotype** (*observable traits*) and its **genotype** (*genetic makeup*).

Using these definitions: *PP* represents a homozygous dominant genotype expressing the dominant phenotype; *Pp* represents a heterozygous genotype also expressing the dominant phenotype; and *pp* represents a homozygous recessive genotype expressing the recessive phenotype. Importantly, identical phenotypes need not correspond to identical genotypes.

Organisms displaying a dominant phenotype may be either homozygous dominant (*PP*) or heterozygous (*Pp*). To determine the genotype, a **testcross** is performed by crossing the unknown individual with a homozygous recessive individual (*pp*). If any offspring exhibit the recessive phenotype, the unknown parent must be heterozygous (*Pp*). If all offspring exhibit the dominant phenotype, the unknown parent is most likely homozygous dominant (*PP*).

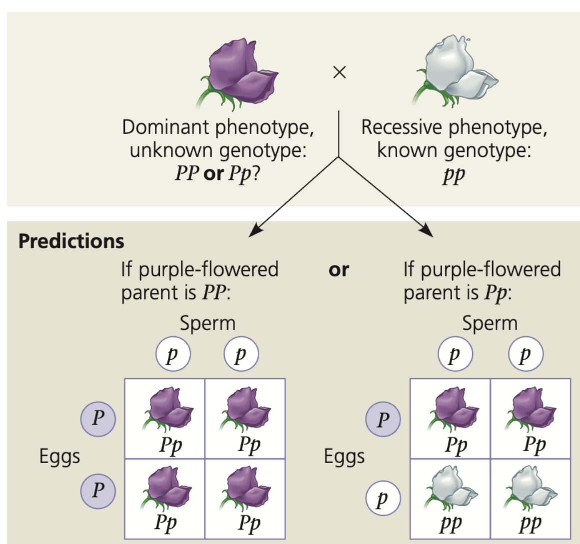


Figure 16: Test cross

Core Concept 2.4: Law of Segregation

The **law of segregation** states that the two alleles for each gene separate during gamete formation, such that each gamete receives only one allele.

The law of segregation can be illustrated using a **Punnett square** (*Figure 17*), which displays all possible allele combinations in offspring given parental **genotypes** (*Core Concept 2.3*). Each cell in the grid represents an equally likely fertilization event, and the relative frequencies of different genotypes can be predicted from the number of cells showing each combination.

The predictive power of the law of segregation rests on probability. While the Punnett square shows all possible outcomes, each individual offspring results from a single random fertilization.

	R	r
R	**RR**	**Rr**
R	**RR**	**Rr**

Figure 17: Punnett square showing cross between heterozygous father (*Rr*) and homozygous dominant mother (*RR*)

Core Concept 2.5: Law of Independent Assortment & Dihybrid Crosses

Mendel's **law of independent assortment** states that alleles of different genes assort independently during gamete formation.

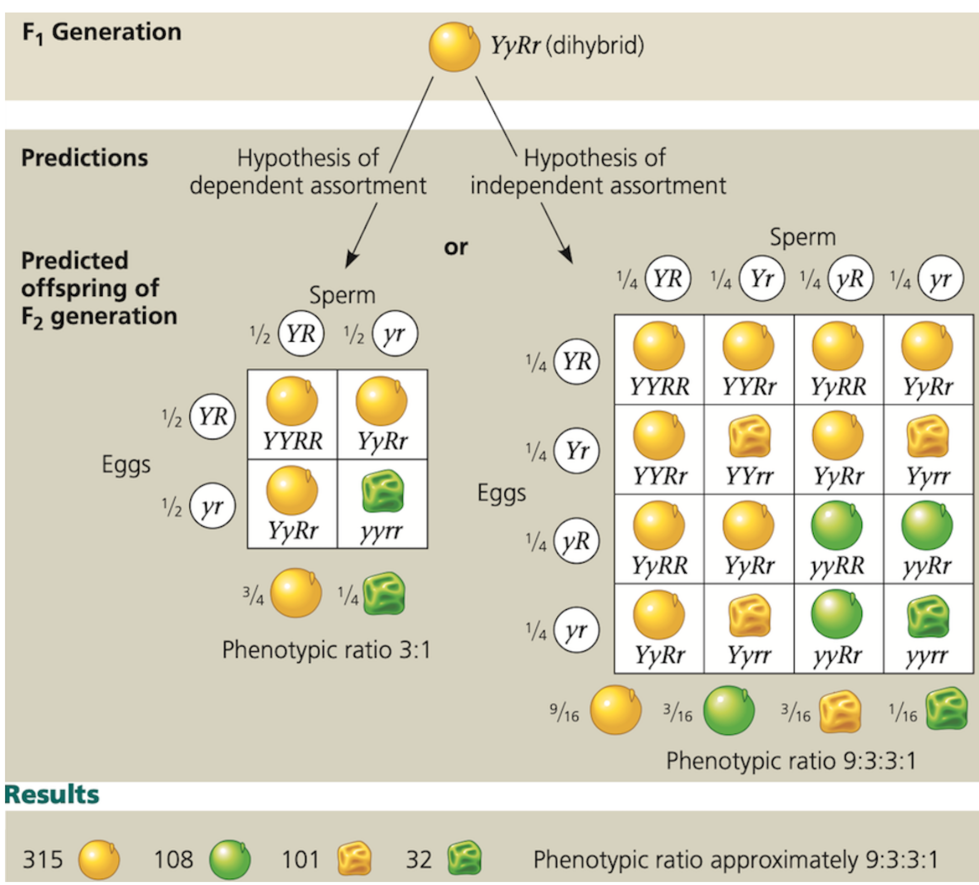


Figure 18: Dihybrid cross examining seed color and shape

In order to derive this law, Mendel also examined two characters simultaneously, such as seed color (*yellow/Y or green/y*) and seed shape (*round/R or wrinkled/r*). Crossing true-breeding parents differing in both traits (*YYRR × yyrr*) produced F1 **dihybrids** heterozygous for both characters (*YyRr*).

If genes were inherited together (*dependent assortment*), F2 offspring would exhibit a 3:1 phenotypic ratio. However, independent assortment produces the characteristic 9:3:3:1 ratio observed in **dihybrid crosses**. Mendel's experimental results supported independent assortment.

2.2 Complex Inheritance Patterns

While Mendel established foundational principles for simple traits, many heritable characteristics exhibit more complex patterns, and the genotype-phenotype relationship is often not straightforward.

Core Concept 2.6: Single Gene Abnormal Inheritance Patterns

We first consider non-Mendelian patterns involving a single gene.

(a) **Complete dominance:** In Mendel's classic pea crosses, F1 offspring always resembled one parental variety because one allele exhibited complete dominance over the other.

(b) **Incomplete dominance:** For some genes, neither allele is completely dominant, and F1 hybrids exhibit an intermediate phenotype. For example, crossing true-breeding red and white flowers produces F1 offspring with pink flowers.

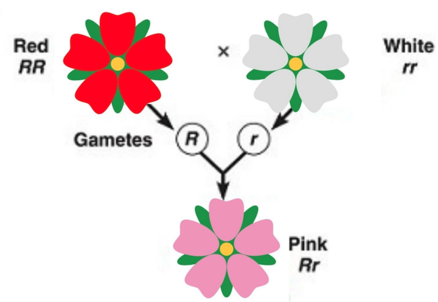


Figure 19: Incomplete dominance of flower color

(c) **Codominance:** Both alleles independently affect the phenotype in distinguishable ways. For example, the human MN blood group is determined by codominant alleles encoding M and N surface molecules on red blood cells. Homozygous individuals express only one molecule type, whereas heterozygous individuals express both.

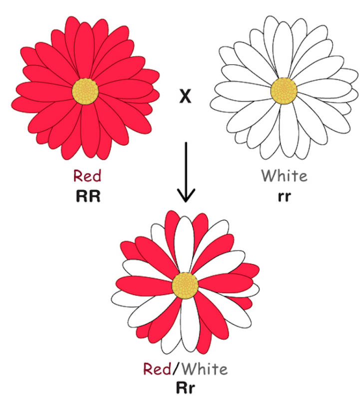
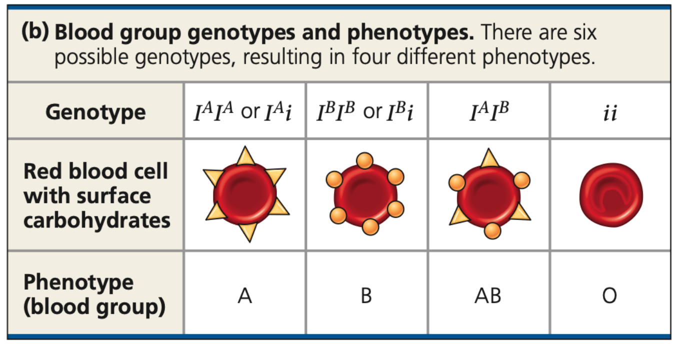


Figure 20: Codominance of flower color

Whether a gene displays incomplete dominance or codominance depends on the level of phenotypic analysis. The heterozygous pink flower in Figure 19, for instance, appears uniformly pink macroscopically (*incomplete dominance*), yet molecular analysis may reveal a mixture of red and white pigment molecules (*codominance*). Classification therefore depends on observational scale.

(d) **Multiple Alleles:** Most genes exist in more than two allelic forms within a population. The human ABO blood group, for instance, is determined by three alleles: I^A, I^B, and i, with each individual possessing any two of these.



(b) Blood group genotypes and phenotypes. There are six possible genotypes, resulting in four different phenotypes.

Genotype	I^AI^A or I^Ai	I^BI^B or I^Bi	I^AI^B	ii
Red blood cell with surface carbohydrates				
Phenotype (blood group)	A	B	AB	O

Figure 21: Multiple alleles of human blood groups

(e) **Pleiotropy:** Single genes may influence multiple phenotypic traits. For instance, the gene determining pea flower color also affects seed coat color (*gray or white*).

Core Concept 2.7: Multi-Gene Abnormal Inheritance Patterns

We now consider non-Mendelian patterns involving multiple genes.

(a) **Epistasis:** The phenotypic expression of one gene modifies the expression of another gene at a different **locus** (*chromosomal location*). In Labrador retrievers, coat color is determined by two genes: one locus controls pigment type (*black B is dominant to brown b*), while a second locus controls pigment deposition (*dominant E allows deposition*). However, homozygous recessive individuals (*ee*) have yellow coats regardless of genotype at the pigment-type locus.

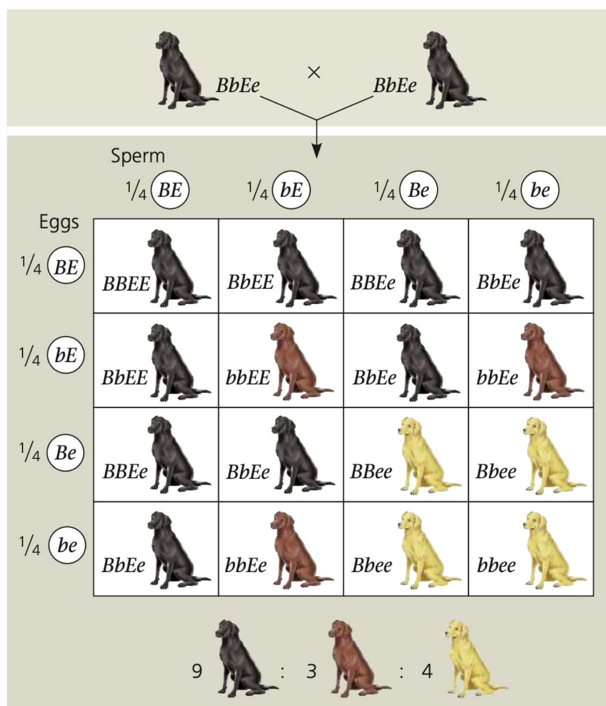


Figure 22: Epistasis of Labrador retriever coat color

(b) **Polygenic Inheritance:** Many characteristics (*e.g., skin color, height*) exhibit continuous variation rather than discrete alternatives. Such **quantitative characters** result from the additive effects of multiple genes. Below is a simplified model of skin color determined by three genes.

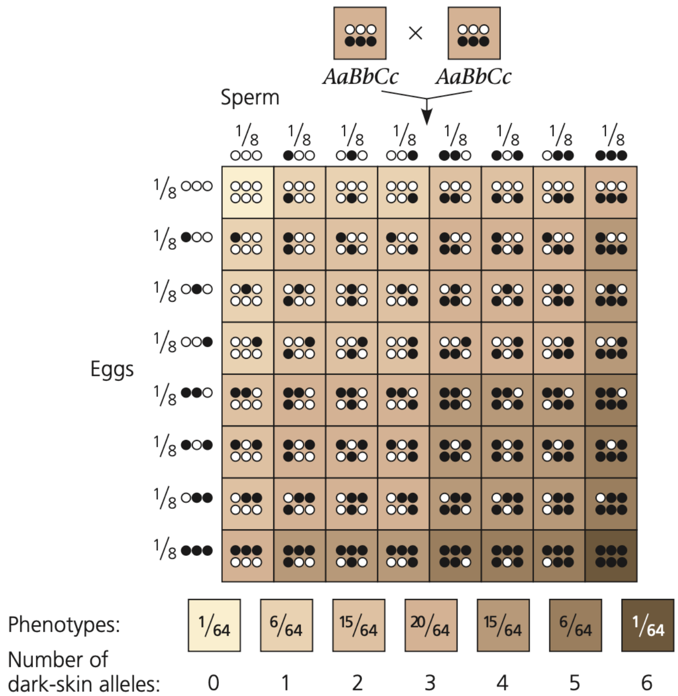


Figure 23: Polygenic inheritance of skin color

Core Concept 2.8: Multifactorial Characteristics

While genotype provides the foundation for phenotype, certain characteristics (*such as skin color*) are also influenced by environmental factors (*e.g., sun exposure*). Such traits are termed **multifactorial**, reflecting the combined influence of multiple genetic and environmental factors on phenotypic expression.

2.3 Use Cases for Mendelian Patterns of Inheritance

Core Concept 2.9: Pedigrees

In lieu of controlled breeding experiments (*infeasible for ethical reasons*), human geneticists analyze inheritance patterns using **pedigrees**, which are family trees that trace the transmission of a particular trait across generations.

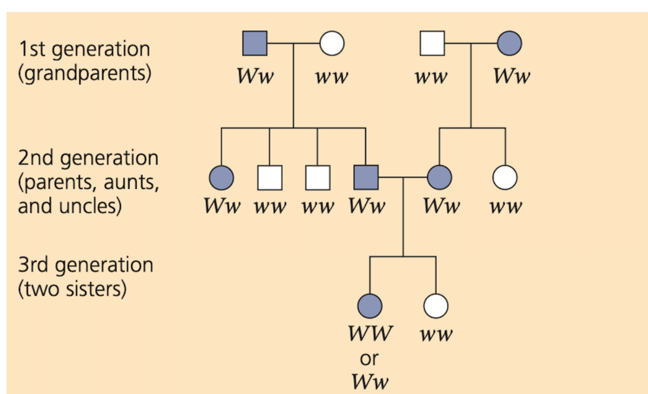


Figure 24: Pedigree analysis of widow's peak (*recessive trait*)

Core Concept 2.10: Pedigree Analysis of Disorders

Pedigree analysis is particularly valuable for studying disease-causing alleles. In **recessive disorders**, heterozygous individuals (*Aa*) exhibit normal phenotypes because one functional allele produces sufficient protein. These **carriers** can transmit the recessive allele to offspring, though only homozygous recessive individuals (*aa*) express the disorder. Most affected individuals are born to phenotypically normal carrier parents, and lethal recessive alleles remain rare because affected individuals typically do not reproduce.

In **dominant disorders**, heterozygous individuals are phenotypically affected. **Achondroplasia**, a form of dwarfism caused by a dominant allele, persists in populations because the condition does not prevent reproduction; unaffected individuals are homozygous recessive. Dominant lethal alleles are generally rare because affected individuals die before reproducing. However, such alleles can persist if symptoms manifest after reproductive age, as in **Huntington's disease**, a neurodegenerative disorder with typical onset in mid-adulthood.

Core Concept 2.11: Multifactorial Diseases

While the preceding discussion focused on single-locus Mendelian disorders, many common diseases have a **multifactorial** basis involving both genetic predisposition and environmental influences. Susceptibility to cardiovascular disease or stroke, for instance, depends on both genotype and lifestyle factors (*smoking, exercise, etc.*).

3 The Chromosomal Basis of Inheritance

3.1 Sex-Linked Genes

When Mendel formulated the laws of inheritance in 1866, the physical basis of heredity remained unknown. The **chromosome theory of inheritance**, proposed in 1902, established that Mendelian genes occupy specific loci on chromosomes and that chromosomal behavior during meiosis accounts for observed inheritance patterns.

Core Concept 3.1: Morgan's Fruit Fly Experiment

The first compelling evidence linking a specific gene to a specific chromosome came from Thomas Hunt Morgan's early 1900s work on eye color inheritance in *Drosophila melanogaster*.

The prevalent phenotype in wild populations is red eye color, termed the **wild-type** (w^+). After two years of breeding, Morgan discovered a single white-eyed male, representing the **mutant phenotype** (w).

Crossing the white-eyed male with red-eyed females produced all red-eyed F1 offspring, indicating dominance of the wild-type allele. F1 intercrosses yielded an overall 3:1 red-to-white F2 ratio. However, the white-eye phenotype appeared exclusively in males—approximately half of F2 males were white-eyed while all females were red-eyed. This sex-specific pattern led Morgan to conclude that the eye-color gene resides on the X chromosome.

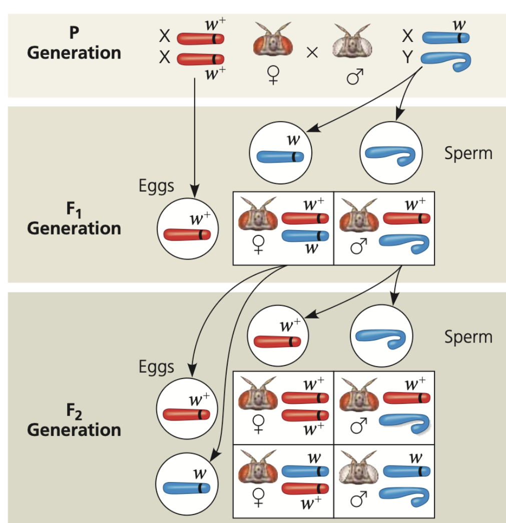


Figure 25: Morgan's fruit fly experiment crosses

Because all F1 offspring exhibited the wild-type phenotype, F1 males must have inherited a wild-type allele on their single X chromosome, while F1 females must have been heterozygous (w^+w). The appearance of white-eyed F2 males confirmed this: males inherit their X chromosome exclusively from their mother, so white-eyed males received the mutant allele from heterozygous F1 females, establishing them as carriers.

Core Concept 3.2: Sex Chromosomes

Humans and other mammals possess two sex chromosomes, X and Y. Females are typically XX, while males are XY. The Y chromosome is considerably smaller than the X, with homology restricted to short regions at their termini. These homologous regions enable X-Y pairing and proper segregation during male meiosis.

During gamete formation, sex chromosomes segregate such that all eggs carry an X chromosome, while sperm carry either X or Y. Fertilization by X-bearing sperm produces XX (*female*) zygotes; Y-bearing sperm produces XY (*male*) zygotes. This mechanism yields an approximate 1:1 sex ratio. However, the mammalian X-Y system represents only one of several sex-determination mechanisms in nature (*Figure 26*).

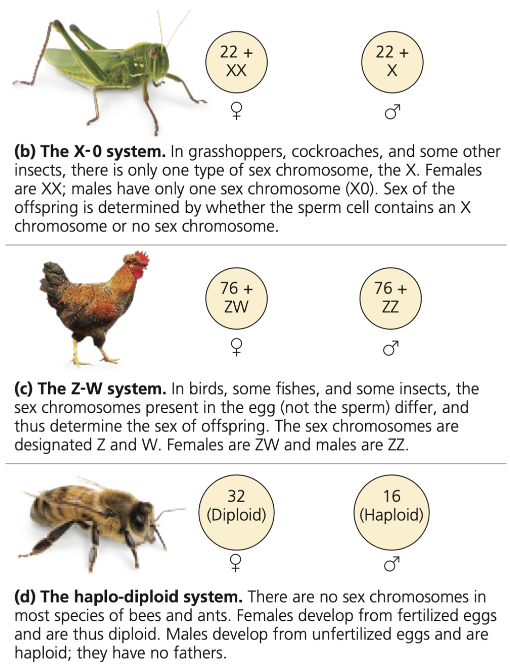


Figure 26: Non-mammalian sex-determination mechanisms

Core Concept 3.3: Sex-Linked Genes

Genes located on sex chromosomes are termed **sex-linked genes**. Those on the X chromosome are **X-linked**; those on the Y are **Y-linked**. The human X chromosome contains approximately 1,100 genes, while the Y chromosome contains roughly 78 genes encoding approximately 25 proteins, many involved in testis development and sperm production. Because the Y chromosome is transmitted essentially unchanged from father to son and carries relatively few genes, Y-linked inheritance is comparatively rare.

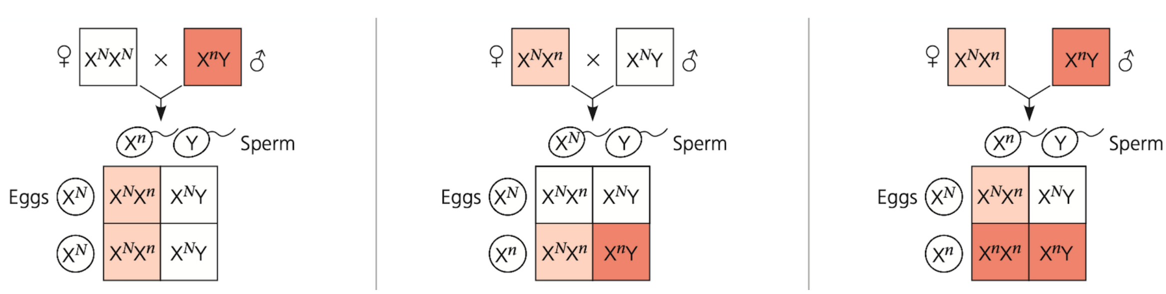


Figure 27: Inheritance patterns of sex-linked genes

X-linked genes exhibit the inheritance pattern Morgan first observed in *Drosophila*. Fathers transmit their single X chromosome—and all X-linked alleles—exclusively to daughters. Mothers, possessing two X chromosomes, can transmit X-linked alleles to offspring of either sex.

Phenotypic expression of X-linked traits in females depends on allele dominance. Because males possess only one X chromosome, the terms homozygous and heterozygous do not apply. Instead, males are **hemizygous** for X-linked genes, expressing whichever allele is present on their single maternal X chromosome.

Core Concept 3.4: X Inactivation and Barr Bodies

Although female mammals inherit two X chromosomes, one undergoes **inactivation** early in embryonic development, ensuring dosage compensation between sexes for most X-linked gene products. The inactivated X chromosome condenses into a **Barr body** at the nuclear periphery, rendering most of its genes transcriptionally silent. In germline cells, the inactive X is reactivated prior to meiosis, ensuring each egg contains one active X chromosome.

X inactivation renders females genetic **mosaics** composed of two cell populations: those expressing the maternal X and those expressing the paternal X. Once an X chromosome is inactivated in a cell, all mitotic descendants maintain the same inactivation pattern. Consequently, females heterozygous for an X-linked trait exhibit a mosaic phenotype with patches of cells expressing alternate alleles.

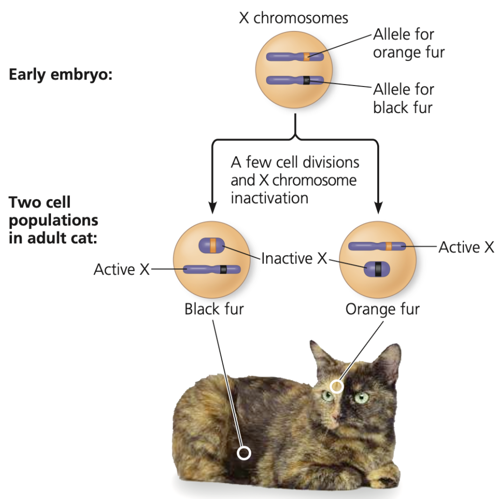


Figure 28: Fur color genetic mosaic in female cat due to Barr body development

This mosaicism is evident in women heterozygous for X-linked mutations affecting sweat gland development, who exhibit distinct patches of normal skin interspersed with regions lacking sweat glands.

3.2 Linked Genes

Core Concept 3.5: Gene Linkage

Genes located proximally on the same chromosome tend to be inherited together; such genes are **genetically linked**. Consider Morgan's experiment crossing true-breeding double wild-type flies with true-breeding double mutant flies.

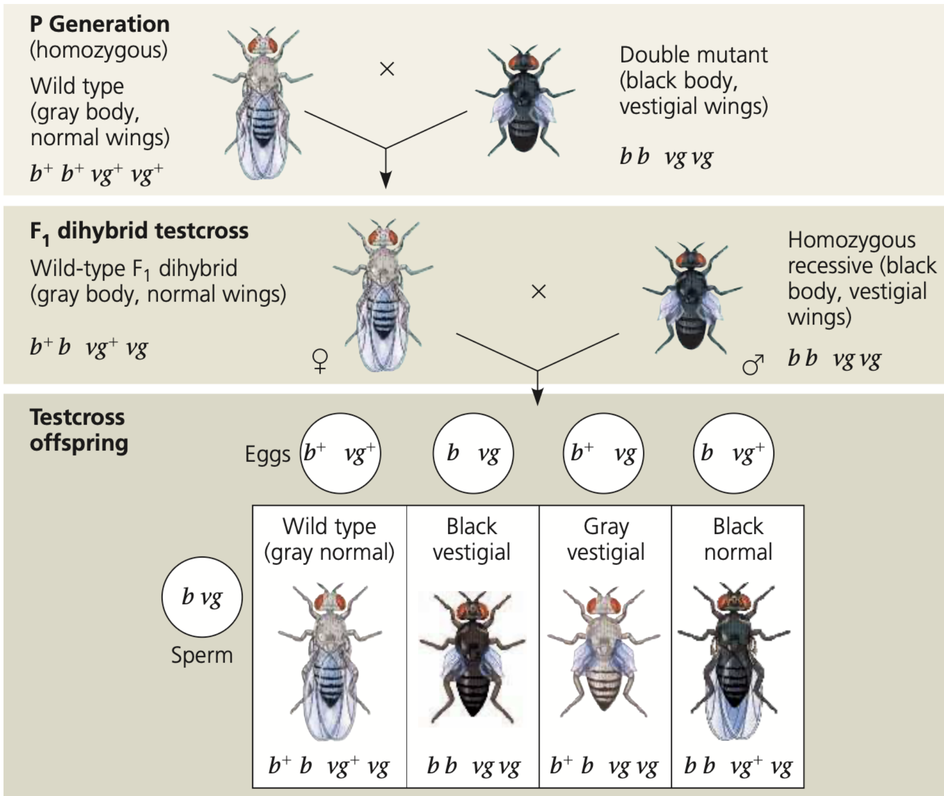


Figure 29: True-breeding double wild-type and mutant cross showing evidence of linkage between genes

If genes reside on different chromosomes, independent assortment predicts a 1:1:1:1 testcross ratio. Complete linkage (*genes on the same chromosome with no recombination*) predicts a 1:1:0:0 ratio, yielding only parental phenotypes. Morgan's actual data—965:944:206:185—fell between these extremes. This intermediate pattern results from **crossing over**, though the molecular mechanism remained unknown at the time.

Core Concept 3.6: Parental Types vs Recombinants

Genetic recombination produces offspring with trait combinations differing from either P generation parent. Trait combinations matching the original parental combinations are **parental types**; those differing are **recombinant types** (*or **recombinants***).

Core Concept 3.7: Linkage Maps

Crossing over occurs randomly along homologous chromosomes during prophase I. The probability of crossover between two genes increases with their physical separation. Thus, more distantly spaced genes exhibit higher recombination frequencies, approaching 50%—the value expected for independent assortment on different chromosomes.

Using recombination frequencies, Alfred H. Sturtevant, Morgan's student, constructed the first **linkage maps** by assigning relative positions to genes on chromosomes based on *Drosophila* recombination data (*Figure 30*).

Genetic linkage is not binary but continuous, determined by physical distance between genes: closer genes are more tightly linked and exhibit lower recombination frequencies.

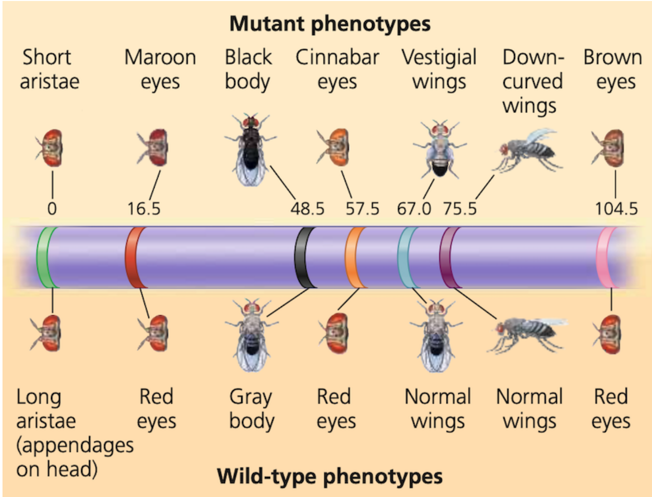


Figure 30: Linkage map

3.3 Chromosomal Disorders

Thus far, discussion has focused on single-gene alterations (*Core Concept 2.10*). However, large-scale chromosomal abnormalities can also profoundly affect phenotype and may cause miscarriage or developmental disorders.

Core Concept 3.8: Abnormal Chromosome Count

Abnormal chromosome numbers typically arise from **nondisjunction**, wherein homologous chromosomes fail to separate during meiosis I or sister chromatids fail to separate during meiosis II, producing gametes with abnormal chromosome counts.

Fertilization involving an **aneuploid** (*abnormal chromosome count*) gamete produces an aneuploid zygote. Loss of one chromosome yields a **monosomic** zygote ($2n-1$); gain of one produces a **trisomic** zygote ($2n+1$). This chromosomal abnormality propagates to all somatic cells through mitotic divisions.

In contrast, **polyploidy** describes organisms possessing more than two complete chromosome sets, including *triploidy* ($3n$) and *tetraploidy* ($4n$).

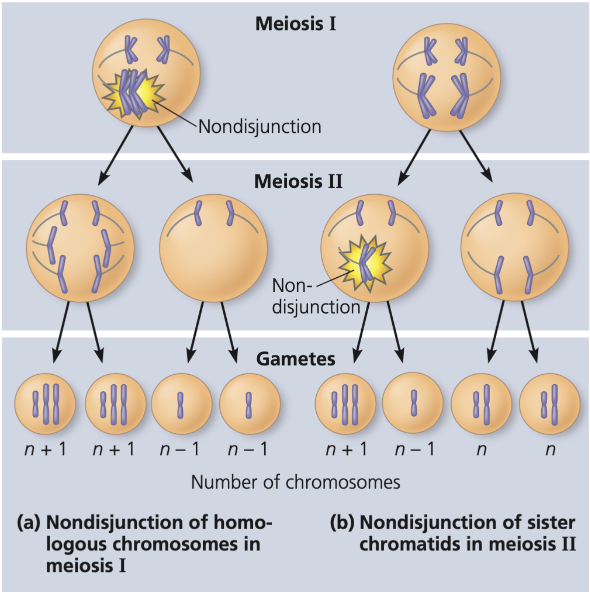


Figure 31: Nondisjunction

Core Concept 3.9: Abnormal Chromosome Structure

Beyond numerical abnormalities, meiotic errors or exposure to mutagens such as radiation can alter chromosome structure. Structural abnormalities comprise four major categories.

(a) **Deletion**: Loss of a chromosomal fragment, eliminating specific genes.

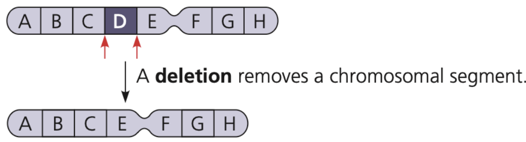


Figure 32: Chromosomal fragment deletion

(b) **Duplication**: Attachment of a chromosomal fragment as an additional segment to a sister or non-sister chromatid, producing repeated gene sequences.

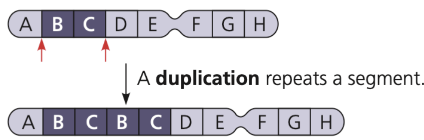


Figure 33: Chromosomal fragment duplication

(c) **Inversion**: Reattachment of a chromosomal fragment in reverse orientation, inverting gene order within that segment.

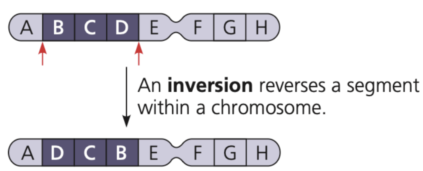


Figure 34: Chromosomal fragment inversion

(d) **Translocation**: Attachment of a chromosomal fragment to a nonhomologous chromosome. Reciprocal translocation involves segment exchange between two nonhomologous chromosomes.

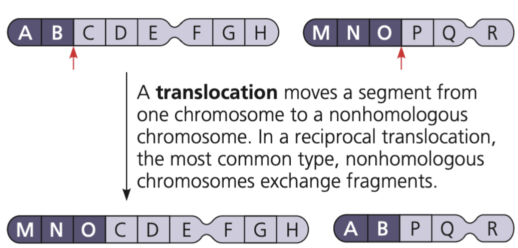


Figure 35: Chromosomal fragment translocation

Core Concept 3.10: Notable Chromosomal Abnormalities

Certain aneuploidies produce recognizable clinical syndromes in humans.

(a) **Down syndrome** (*trisomy 21*) results from an extra copy of chromosome 21.

Sex chromosome aneuploidies typically cause less severe phenotypic disruption than autosomal aneuploidies. This reflects the Y chromosome's limited gene content and the inactivation of additional X chromosomes as Barr bodies.

(b) **Klinefelter syndrome** (*XXY*) affects phenotypic males who may exhibit small testes, reduced testosterone, tall stature, decreased muscle mass, sparse body hair, and gynecomastia. Many affected individuals are infertile.

(c) **Trisomy X** (*XXX*) females are generally healthy with few unusual physical features beyond slightly above-average height, though learning disabilities and mild developmental delays occur at increased frequency.

(d) **Turner syndrome** (*monosomy X, XO*) represents the only viable human monosomy. Affected individuals are phenotypically female but typically sterile due to ovarian dysgenesis and incomplete sexual maturation.

3.4 Exceptions to Mendelian Inheritance

Beyond chromosomal abnormalities, two naturally occurring phenomena deviate from classical Mendelian inheritance.

Core Concept 3.11: Genomic Imprinting

Genomic imprinting describes parent-of-origin-dependent allele expression. For imprinted genes, either the maternal or paternal allele is consistently silenced while the other is expressed. This parent-specific expression pattern is regulated by epigenetic modifications such as DNA methylation established during gamete formation. For example, only the paternal allele of the mouse *insulin-like growth factor 2* (*Igf2*) gene is expressed.

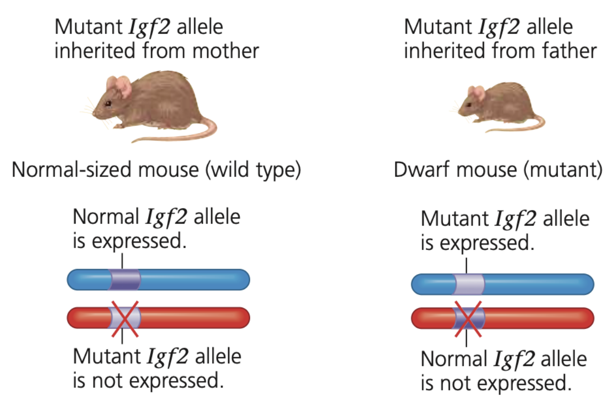


Figure 36: Genomic imprinting of *Igf2* gene in mice

Core Concept 3.12: Inheritance of Organelle Genes

Extranuclear inheritance involves transmission of genes located in organellar genomes—the circular DNA molecules in mitochondria and (*in plants*) chloroplasts.

In most animals, zygotes receive nearly all cytoplasm and organelles from the egg; sperm contribute primarily nuclear chromosomes. Consequently, mitochondrial DNA (*and chloroplast DNA in plants*) exhibits **maternal inheritance**.